

HW2.

1. Prove Gauss-Markov Theorem

We want to show $\text{Var}(\hat{\alpha}^T \hat{\beta}) \leq \text{Var}(C^T y)$, where $C^T y$ is any unbiased estimator for $\alpha^T \beta$.

let $y = X\beta + \varepsilon$ $E(\varepsilon) = 0$ $\text{Var}(\varepsilon) = \sigma^2 I$, estimator for $\alpha^T \beta$

① OLS: $\hat{\beta} = (X^T X)^{-1} X^T y$ $\alpha^T \hat{\beta} = \alpha^T (X^T X)^{-1} X^T y$ let $C_0 = X(X^T X)^{-1} \alpha$

$$\Rightarrow C_0^T y = \alpha^T (X^T X)^{-1} X^T y = \alpha^T \hat{\beta}$$

$$E(C_0^T \hat{\beta}) = E(C_0^T (X^T X)^{-1} X^T y) \quad \because E(y) = E(X\beta + \varepsilon) = X\beta + 0 = X\beta$$

$$\Rightarrow E(\alpha^T \hat{\beta}) = \alpha^T C_0^T X \beta = \alpha^T \beta$$

Therefore, it is unbiased.

② Let another estimator $\theta' = C^T y$ $E(\theta') = \alpha^T \beta$ $E(\theta') = C^T E(y) = C^T X \beta$

$$\Rightarrow C^T X = \alpha^T$$

Therefore, $X^T C = \alpha$

$$\because \text{Var}(y) = \sigma^2 I \quad \Rightarrow \quad \text{Var}(C^T y) = C^T (\sigma^2 I) C = \sigma^2 C^T C$$
$$\text{Var}(C_0^T \hat{\beta}) = \text{Var}(C_0^T y) = \sigma^2 C_0^T C_0$$

Therefore we need to show $C^T C \geq C_0^T C_0$.

③ let $d = C - C_0 \Rightarrow C = C_0 + d$ $X^T C = \alpha$ $C_0 = X(X^T X)^{-1} \alpha$

$$\Rightarrow X^T C_0 = X^T X (X^T X)^{-1} \alpha = \alpha$$

$$X^T d = X^T (C - C_0) = X^T C - X^T C_0 = \alpha - \alpha = 0$$

Therefore $X^T d = 0$

$$\textcircled{4} \quad C^T C = (C_0 + d)^T (C_0 + d) = C_0^T C_0 + C_0^T d + d^T d$$

$$C_0^T d = \alpha^T (X^T X)^{-1} X^T d = \alpha^T (X^T X)^{-1} \cdot 0 = 0$$

$$\Rightarrow C^T C = C_0^T C_0 + d^T d$$

$\because d^T d > 0$

$$\Rightarrow C^T C > C_0^T C_0 \Rightarrow \sigma^2 C^T C > \sigma^2 C_0^T C_0$$

Therefore, $\text{Var}(C^T y) \geq \text{Var}(\alpha^T \beta)$ OLS is BLUE

2.

$$\text{Let } X = [x_0 \ x_1 \ \dots \ x_p] \quad x_i^T x_j = 0 \text{ if } i \neq j \quad y = X\beta + \varepsilon$$

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

$$\Rightarrow X^T X = \begin{bmatrix} x_0^T x_0 & 0 & \dots & 0 \\ 0 & x_1^T x_1 & & \\ \vdots & & \ddots & \\ 0 & \dots & & x_p^T x_p \end{bmatrix} \Rightarrow (X^T X)^{-1} = \begin{bmatrix} \frac{1}{x_0^T x_0} & 0 & \dots & 0 \\ 0 & \frac{1}{x_1^T x_1} & & \\ \vdots & & \ddots & \\ 0 & \dots & & \frac{1}{x_p^T x_p} \end{bmatrix}$$

$$\Rightarrow X^T y = \begin{bmatrix} x_0^T y \\ x_1^T y \\ \vdots \\ x_p^T y \end{bmatrix}$$

$$\hat{\beta} = \begin{bmatrix} \frac{1}{x_0^T x_0} & & & \\ & \frac{1}{x_1^T x_1} & & \\ & & \ddots & \\ & & & \frac{1}{x_p^T x_p} \end{bmatrix} \begin{bmatrix} x_0^T y \\ x_1^T y \\ \vdots \\ x_p^T y \end{bmatrix} = \begin{bmatrix} \frac{x_0^T y}{x_0^T x_0} & \frac{x_1^T y}{x_1^T x_1} & \dots & \frac{x_p^T y}{x_p^T x_p} \end{bmatrix}^T$$

$$\Rightarrow \text{therefore, } \hat{\beta}_j = \frac{x_j^T y}{x_j^T x_j}$$

3.

a) show that $\beta_{\text{OLS}} = V \Sigma^{-1} U^T y$ is a solution to the normal equations.

$$\text{We know that } X = U \Sigma V^T \quad X^T X \beta = X^T y \quad U^T U = I_r \quad V^T V = I_r$$

$$\Rightarrow X^T X = (U \Sigma V^T)^T (U \Sigma V^T) = V \Sigma^T U^T U \Sigma V^T = V \Sigma^2 V^T$$

$$X^T X \beta_{\text{OLS}} = (V \Sigma^2 V^T) (V \Sigma^{-1} U^T y) = V \Sigma U^T y$$

$$X^T y = (U \Sigma V^T)^T y = V \Sigma U^T y$$

$$\Rightarrow \text{Therefore, } X^T X \beta_{\text{OLS}} = X^T y$$

b) to show that $\|\beta\| \geq \|\beta_{\text{OLS}}\|$

$$\text{We have } X^T X \beta = X^T y$$

$$X^T X \beta_{\text{OLS}} = X^T y$$

$$\Rightarrow X^T X (\beta - \beta_{\text{OLS}}) = 0$$

$$\text{Let } b = \beta - \beta_{\text{OLS}} \Rightarrow X^T X b = 0$$

$$\Rightarrow b^T X^T X b = b^T \cdot 0 = 0$$

$$\|Xb\|^2 = 0 \Rightarrow Xb = 0 \quad b \in \text{Null}(X)$$

Therefore, $\beta = \beta_{\text{OLS}} + b \quad b \in \text{Null}(X)$

$$\because Xb = 0 \quad X = U \Sigma V^T$$

$$\therefore U \Sigma V^T b = 0 \quad V^T b = 0$$

$$\therefore \beta_{\text{OLS}} = V \Sigma^{-1} U^T y$$

$$\Rightarrow \beta_{\text{OLS}}^T b = (U^T y)^T \Sigma^{-1} V^T b = 0$$

Therefore, $\beta_{\text{OLS}} \perp b$

$$\therefore \beta = \beta_{\text{OLS}} + b$$

$$\therefore \|\beta\|^2 = \|\beta_{\text{OLS}}\|^2 + \|b\|^2$$

$$\therefore \|\beta\|^2 \geq \|\beta_{\text{OLS}}\|^2$$

Therefore, $\|\beta\| \geq \|\beta_{\text{OLS}}\|$ when $b=0$ $\beta = \beta_{\text{OLS}}$.

(c)

if $V \Sigma^+ U^T$ is the pseudo-inverse of X

we need to show

- ① $X X^+ X = X$
- ② $X^+ X X^+ = X^+$
- ③ $(X X^+)^T = X X^+$
- ④ $(X^+ X)^T = X^+ X$

$$\begin{aligned} \textcircled{1} & (U \Sigma V^T)(V \Sigma^+ U^T)(U \Sigma V^T) \\ &= U \Sigma \Sigma^+ \Sigma V^T = U \Sigma V^T = X \end{aligned}$$

$$\begin{aligned} \textcircled{2} & (V \Sigma^+ U^T)(U \Sigma V^T)(V \Sigma^+ U^T) \\ &= V \Sigma^+ \Sigma \Sigma^+ U^T = V \Sigma^+ U^T = X^+ \end{aligned}$$

$$\begin{aligned} \textcircled{3} & ((U \Sigma V^T)(V \Sigma^+ U^T))^T = (U U^T)^T = U U^T \\ & (U \Sigma V^T)(V \Sigma^+ U^T) = U U^T \end{aligned}$$

$$\begin{aligned} \textcircled{4} & (V \Sigma^+ U^T)(U \Sigma V^T)^T = V V^T \\ & (V \Sigma^+ U^T)(U \Sigma V^T) = V V^T \end{aligned}$$

$$\Rightarrow \text{Therefore } X^+ = V \Sigma^+ U^T$$